

C++ Interview Preparation (25 LPA)

C++ Fundamentals - Interview PDF Guide (25 LPA Focus)

1. Basics: Syntax, keywords, scope resolution (::), namespaces
2. Data Types & Storage: static, extern, auto, register, volatile, mutable
3. Memory Management: new/delete, malloc/free, stack vs heap
4. OOP: Classes, Inheritance, Encapsulation, Polymorphism
5. Constructors/Destructors: Rule of 3/5/0, copy/move constructors
6. Operator Overloading: +, [], <<, = operators
7. Function Overloading vs Overriding: Static vs Dynamic binding
8. Virtual Functions: Pure virtual, abstract classes, vtables
9. Templates: Function, class, specialization
10. Smart Pointers: unique_ptr, shared_ptr, weak_ptr
11. Exception Handling: try, catch, throw, stack unwinding
12. STL: vector, map, set, priority_queue, iterators, algorithms
13. Lambda Functions: Syntax, capture lists, mutability
14. Type Casting: static_cast, dynamic_cast, const_cast, reinterpret_cast
15. Const Correctness: const with methods, pointers, references
16. References vs Pointers: nullability, reassignment, use cases
17. Multithreading: std::thread, mutex, deadlocks
18. Move Semantics: std::move, rvalue references, efficiency
19. Memory Leaks & Tools: Valgrind, AddressSanitizer
20. Inline Functions & Macros: Pros/cons, differences
21. Static Members: class-level variables and functions
22. Compilation Model: headers, ODR, include guards
23. Design Patterns: Singleton, Factory (optional)
24. Undefined Behavior: dangling pointers, uninitialized memory
25. Modern C++ Features: auto, decltype, nullptr, constexpr

MCQ-Style Prep - Sample Questions

Q1. Which of the following deletes memory allocated with new?

- A) free()
- B) delete

C++ Interview Preparation (25 LPA)

C) delete[]

D) Both B and C

Ans: D

Q2. Which feature of OOP allows function overriding?

A) Encapsulation

B) Inheritance

C) Abstraction

D) Polymorphism

Ans: D

Q3. What is the output of `static_cast<int>(3.14)`?

A) 3

B) 3.14

C) 4

D) Error

Ans: A

Q4. What does `shared_ptr` use internally to manage the count?

A) Atomic int

B) Raw pointer

C) Reference count

D) Mutex

Ans: C

Q5. Which of these is NOT a smart pointer in C++?

A) `unique_ptr`

B) `shared_ptr`

C) `raw_ptr`

D) `weak_ptr`

Ans: C